# APEX INSTITUTE OF TECHNOLOGY & SCIENCE (AUTONOMOUS)

Department of Computer Science & Engineering
Curriculum & Detailed Syllabi (Regulations 2024 - AICTE Model Framework)

## SEMESTER III

### *Course Code: PCC-CS301 : Data Structures and Algorithms*

**Credits: 4.0** (L: 3, T: 1, P: 0) | Category: Professional Core

**Course Outcomes:**
CO1: Understand time and space complexity tradeoffs for linear and non-linear data structures.
CO2: Implement binary search trees, AVL trees and priority queues for fast retrieval.
CO3: Analyze graph traversal techniques (BFS, DFS) and shortest path algorithms.
CO4: Design efficient dynamic memory structures and hash collision resolution systems.

**Syllabus Units:**
**UNIT I: Linear Data Structures (8 Hours)** - Arrays, Stacks, Queues, Linked Lists, Circular Lists, Infix to Postfix conversion.
**UNIT II: Trees and Search Structures (9 Hours)** - Binary Trees, Binary Search Trees, AVL Trees, B-Trees, Traversals and balancing.
**UNIT III: Graph Algorithms (9 Hours)** - Adjacency matrix, Breadth First Search, Depth First Search, Dijkstra's algorithm, Minimum Spanning Trees.
**UNIT IV: Sorting and Hashing (8 Hours)** - Merge Sort, Quick Sort, Heap Sort, Hash tables, Open addressing, Chaining strategies.
**TEXT BOOKS:** 1. Thomas H. Cormen, Introduction to Algorithms, MIT Press. 2. Mark Allen Weiss, Data Structures & Algorithm Analysis.

### *Course Code: PCC-CS302 : Database Management Systems*

**Credits: 4.0** (L: 3, T: 1, P: 0) | Category: Professional Core

**Course Outcomes:**
CO1: Formulate entity relationship models and relational schema mappings.
CO2: Apply relational calculus, algebra and complex SQL queries with joins and subqueries.
CO3: Analyze schema normalization up to BCNF eliminating anomalies.
CO4: Evaluate transaction ACID properties, concurrency control and serializability.

**Syllabus Units:**
**UNIT I: Database Architecture & ER Model (8 Hours)** - File systems vs DBMS, Data models, ER diagrams, Constraints.
**UNIT II: Relational Algebra & SQL (9 Hours)** - Selection, Projection, Joins, DDL, DML, Nested queries, Views, Stored procedures.
**UNIT III: Normalization Theory (8 Hours)** - Functional dependencies, 1NF, 2NF, 3NF, Boyce-Codd Normal Form, Multivalued dependencies.
**UNIT IV: Transaction Management (8 Hours)** - Concurrency control, Two-phase locking (2PL), Deadlock handling, Write-Ahead Logging.
**TEXT BOOKS:** 1. Abraham Silberschatz, Database System Concepts, McGraw Hill. 2. Raghu Ramakrishnan, Database Management Systems.

## SEMESTER IV

### *Course Code: PCC-CS401 : Operating Systems*

**Credits: 4.0** (L: 3, T: 1, P: 0) | Category: Professional Core

**Course Outcomes:**
CO1: Understand operating system process scheduling and system call mechanisms.
CO2: Implement concurrency control and synchronization using semaphores and mutexes.
CO3: Analyze virtual memory paging, segmentation and replacement heuristics.

**Syllabus Units:**
**UNIT I: Process Management (9 Hours)** - Process states, Process Control Block, Context switching, CPU scheduling algorithms.
**UNIT II: Synchronization & Deadlocks (9 Hours)** - Critical section, Semaphores, Dining Philosophers, Banker's algorithm.
**UNIT III: Memory Management (8 Hours)** - Paging, Inverted page tables, Demand paging, Page faults, LRU replacement, Thrashing.
**UNIT IV: Storage & File Systems (8 Hours)** - Directory structures, Disk scheduling algorithms (SCAN, C-SCAN), Inodes.
**TEXT BOOKS:** 1. Silberschatz & Galvin, Operating System Concepts, Wiley. 2. Andrew Tanenbaum, Modern Operating Systems.

## SEMESTER V

### *Course Code: PCC-CS501 : Artificial Intelligence & Machine Learning*

**Credits: 4.0** (L: 3, T: 1, P: 0) | Category: Professional Core

**Course Outcomes:**
CO1: Apply search algorithms (A*, Minimax) for intelligent problem solving.
CO2: Implement supervised classification using Decision Trees, SVM and Random Forests.
CO3: Analyze neural network backpropagation and gradient descent optimization.

**Syllabus Units:**
**UNIT I: Search & Intelligent Agents (8 Hours)** - Uninformed & informed search, Heuristic design, A* algorithm, Alpha-beta pruning.
**UNIT II: Supervised Learning (9 Hours)** - Linear regression, Logistic classification, Decision Trees, Support Vector Machines.
**UNIT III: Neural Networks & Deep Learning (9 Hours)** - Multi-layer perceptron, Backpropagation, Activation functions, Regularization.
**TEXT BOOKS:** 1. Stuart Russell and Peter Norvig, AI: A Modern Approach. 2. Tom Mitchell, Machine Learning.